

Machine Learning Assignment Questions

Question 1 - Classification

Dataset: Regression_ML.csv

1. Load the dataset using Pandas.
2. Perform data preprocessing.
3. Split the dataset into training and testing sets.
4. Train the following models:
 - Logistic Regression
 - Decision Tree Classifier
 - Random Forest Classifier
5. Predict the Target column..
6. Compare the models and identify the best model.

Question 2 - Regression

Dataset:Classification_ML.csv

1. Load the dataset using Pandas.
2. Perform data preprocessing.
3. Split the dataset into training and testing sets.
4. Train the following models:
 - Linear Regression
 - Decision Tree Regressor
 - Random Forest Regressor
5. Predict the Target column.
6. Find R2 Score, MAE, MSE, and RMSE for all models.
7. Compare the models and identify the best model.

Question 3 - Advanced Classification Task

1. Create correlation heatmaps and visualize important features.
2. Apply feature scaling using StandardScaler.

Question 4 - Advanced Regression Task

1. Perform feature importance analysis.
2. Create scatter plots between important features and Salary.